

How much the refined +I statistic moves

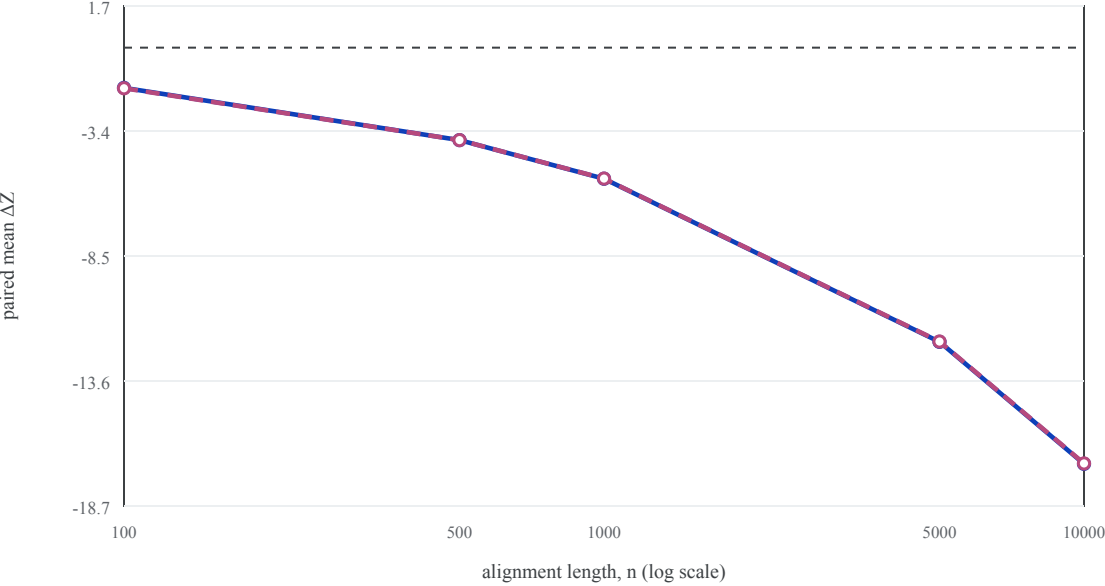
Paired exact-pattern rerun; 20,000 null and 20,000 alternative replicates per cell

+I+G4

+I+R4

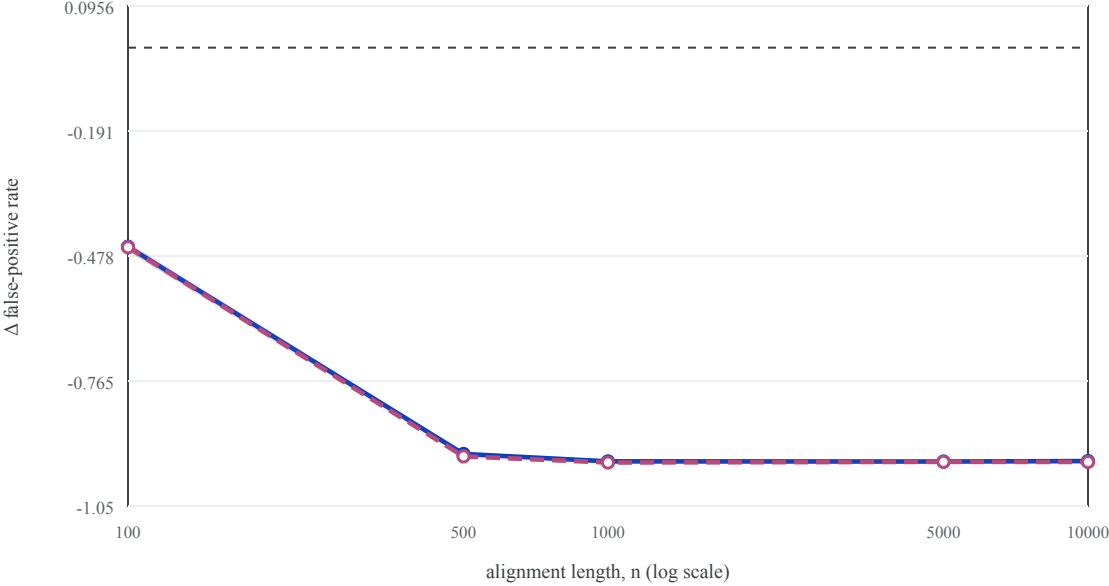
A Null shift in standardized statistic

t = 10,000; refined minus legacy on the same null alignments



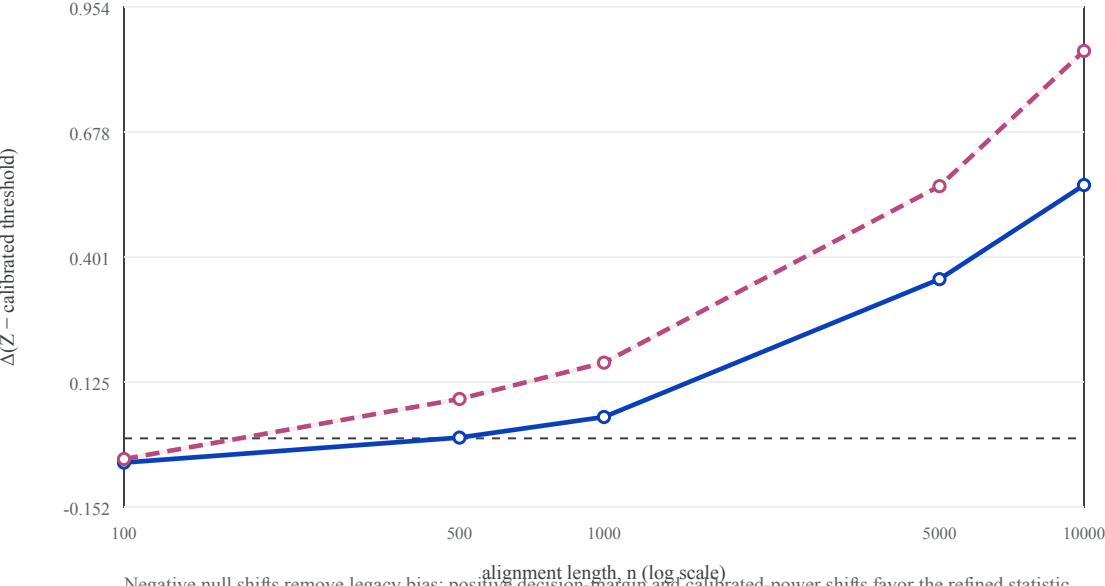
B Nominal false-positive shift

t = 10,000; refined minus legacy rejection fraction



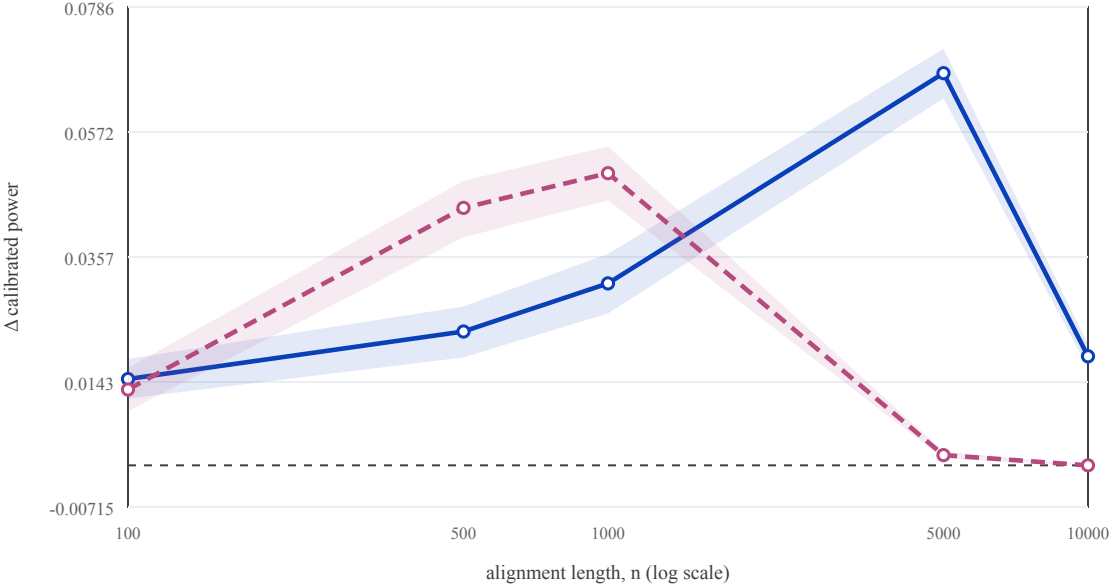
C Calibrated decision-margin shift

+I+G4 at t = 50; +I+R4 at t = 20



D Calibrated-power shift

+I+G4 at t = 50; +I+R4 at t = 20



Negative null shifts remove legacy bias; positive decision-margin and calibrated-power shifts favor the refined statistic.